

Daniel Saromo-Mori

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👤 DanielSaromo

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🎓 Google Scholar

Education

Czech Technical University in Prague

Prague, Czechia

M.Sc. Cybernetics and Robotics, AI Specialization — Currently Enrolled

Sep 2024 – Present

- Awarded merit scholarship for excellent study results in Winter Semester AY 2024/2025; 1st in Czechia (QS Engineering 2025)

Politecnico di Milano

Milan, Italy

M.Sc. Automation and Control Engineering — Weighted Mark: 25.6/30

Sep 2022 – Feb 2024

- Studied at institution ranked 13th worldwide (QS Engineering 2022); won 1st place at Pitch Competition 2023, Entrepreneurship Club POLIMI

Pontificia Universidad Católica del Perú (PUCP)

Lima, Peru

B.Sc. Mechatronics Engineering & Professional Degree

Mar 2014 – Nov 2020

- Graduated in top 6.66% of the Faculty of Science and Engineering (6th of 32 mechatronics graduates)
- Thesis: “Intelligent spider robot for detecting anti-personnel metallic landmines in uneven terrain”
- Earned Best Bachelor’s Thesis award and Innovation Recognition Award at IMECE 2019 (world finals, USA)
- Received Extraordinary Support Funding for Research Thesis; degree thesis rated *outstanding* unanimously

Scientific Publications — 200+ Citations (Google Scholar)

[C4] Valdenegro-Toro, M. and Saromo, D. “A Deeper Look into Aleatoric and Epistemic Uncertainty Disentanglement,” *LXCV Workshop at CVPR 2022*. New Orleans, USA. **200 citations**. Oral + poster.

[C3] Bravo, L., Saromo, D., and Villota, E. “Smart Insole Sensor for vGRF Measurement,” *9th Int’l Symposium on Sensor Science*. Warsaw, Poland. 2022.

[C2] Saromo, D., Bravo, L., and Villota, E. “Smart Sensor Calibration with Auto-Rotating Perceptrons,” *LXAI Workshop at ICML 2020*. Vienna, Austria. Oral + poster.

[C1] Saromo, D., Villota, E., and Villanueva, E. “Auto-Rotating Perceptrons,” *LXAI Workshop at NeurIPS 2019*. Vancouver, Canada. Oral + poster.

[T1] Saromo, D. “Intelligent spider robot for detecting anti-personnel metallic landmines in uneven terrain,” *PUCP*. Lima, Peru. 2020.

[Under Review] “Auto-Rotating Neural Networks: An Alternative Approach for Preventing Vanishing Gradients,” *TMLR*.

Professional & Research Experience

Siemens

Prague, Czechia

Application Engineer, IT/OT Integration (Working Student / DPČ)

Aug 2025 – Present

- Deliver IT/OT integration solutions at Siemens while concurrently pursuing M.Sc. in Cybernetics and Robotics at CTU Prague

German Research Center for Artificial Intelligence (DFKI)

Bremen, Germany (Remote)

Guest Researcher

Aug 2020 – Dec 2020

- Extended the Auto-Rotating Perceptron (ARP) into a new neural network family (ARNN), implementing dense, recurrent, LSTM, GRU, and convolutional auto-rotating layers
- Validated ARNN against equivalent baselines on GPU clusters; presented results at OAMLS (ACML 2021, Singapore)
- Collaborated with Dr. Matias Valdenegro-Toro on novel architecture design and experimental methodology

PUCP Applied Robotics and Biomechanics Research Group (GIRAB)

Lima, Peru

Research Assistant

Mar 2020 – Jul 2020

- Applied ARP to wearable force sensor calibration, achieving 15x improvement in neural network performance versus standard architectures

- Co-authored peer-reviewed paper presented at ICML 2020 (Vienna); research advisor: Dr. Elizabeth Villota

Teaching Experience — 6+ Years, 24+ Semesters

PUCP Continuing Education Department

Lima, Peru

Lecturer — Specialization Diplomas

Sep 2019 – Present

- Instructed AI for Games across 12 consecutive semesters (2019–2025), training 300+ professionals in applied AI
- Taught Data Analysis Methods for Time Series in Diploma in Data Analytics (2022)

PUCP Center for Advanced Manufacturing Technologies (CETAM)

Lima, Peru

Lecturer

Sep 2020 – Sep 2023

- Delivered ML for Industry (7 semesters) and Python for Data Science (5 semesters) for manufacturing professionals

SENAMHI — Peruvian Government

Lima, Peru

Lecturer

May 2022 – Jun 2022

- Designed and delivered introductory AI and ML curriculum for national meteorological services professionals

PUCP Undergraduate School

Lima, Peru

Teaching Assistant — Faculty of Science and Engineering

Mar 2019 – Dec 2019

- Supported instruction in AI, Machine Learning, and Computer Science Applications

Honors & Awards

- **Merit Scholarship**, CTU Prague — excellent study results (2024)
- **1st Place**, Pitch Competition, Entrepreneurship Club POLIMI, Milan (2023)
- **LXAI Travel Grant** (\$1,860), NeurIPS 2019, Vancouver — oral and poster presenter (2019)
- **Innovation Recognition Award** (\$250), Old Guard 63rd Annual Competition, IMECE World Finals, Utah (2019)
- **ASME Travel Award** (\$1,500), representing PUCP and South America at IMECE, Utah (2019)
- **1st Place + Technical Award** (\$850), Old Guard Oral Competition, ASME E-FEST South America, Lima (2019)

Talks & Presentations — 6 Countries

- **Oct 2024**: ARNN pitch, Dny AI, *Prague* **Jun 2022**: Uncertainty paper, CVPR, *New Orleans*
- **Nov 2021**: ARNN poster, ACML/OAMLS, *Singapore* **Jul 2020**: ARP paper, ICML, *Vienna*
- **Dec 2019**: ARP paper, NeurIPS, *Vancouver* **Nov 2019**: Spider robot, IMECE, *Salt Lake City*

Skills & Additional

AI/ML: PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, OpenAI Gym; Deep Learning (MLP, CNN, ARNN/ARP, DRL), Computer Vision, Transfer Learning

Robotics: ROS-1, Jetson Nano, Embedded Systems (Arduino, Raspberry Pi); TIA Portal (PLC/HMI), LabView

Programming: Python, C, C++, MATLAB, $\LaTeX$; NumPy, Pandas, Matplotlib, Plotly, Git, Linux

CAD/CAE: SolidWorks, Inventor, AutoCAD, EagleCAD, Altium Designer

Languages: Spanish (native), English (proficient), Italian (advanced), French (elementary)